

Miguel de Navascués

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RESEARCH

I am a researcher at the CBGP (*Centre de Biologie pour la Gestion des Populations* - INRAE) in Montpellier. My work focuses on the development and evaluation of statistical methods for analyzing population genetic data to infer evolutionary processes, including genetic drift, gene flow, and selection. Currently, my research is centered on three lines:

1. Simulation-based methods for the **joint inference of demographic and adaptive processes**.
2. **Analysis of temporal population genetic data**, such as data from monitored or experimental populations.
3. **Integrative demographic inference** combining genetic and demographic data (such as capture-recapture data).

Career

2009–present Researcher, INRAE UMR1062 Centre de Biologie pour la Gestion des Populations (Montpellier, France).

2021–2023 Visiting Researcher, Human Evolution, EBC, Uppsala University (Sweden).

2019–2021 Marie Skłodowska-Curie Fellow, Human Evolution, EBC, Uppsala University (Sweden).

2007–2009 Postdoctoral researcher, CNRS UMR7625 Écologie et Évolution (Paris, France).

2006–2007 Visiting researcher, Université Libre de Bruxelles (Belgium).

2006 Postdoctoral researcher, Universidad Politécnica de Madrid & Centro de Investigación Forestal-INIA (Spain).

2001–2005 Doctoral candidate, University of East Anglia (UK).

Projects

I have participated, as coordinator and collaborator, to 21 funded research projects since 2003. Selection of projects below.

- **DevOCGen: Développement et applications de nouveaux outils pour la gestion et la conservation des populations naturelles à partir de données génomiques.** BiodivOc - Biodiversité Occitanie (2022–2025). Rol: Collaborator. PI: S. Boitard & R Leblois.

- **INTROSPEC: Genomic consequences and evolutionary causes of introgression in the late stages of speciation.** ANR AAPG 2019 (2020–2025). Rol: Collaborator. PI: P.-A. Crochet.
- **TimeAdapt: Tracking genetic adaptation of populations using time-series genomic data.** Marie Skłodowska-Curie Individual Fellowships 2017 (2019–2021). [doi:10.5281/zenodo.3267084](https://doi.org/10.5281/zenodo.3267084). Role: Fellow.
- **ABCSelection: Detection of loci under selection from temporal population genomic data through ABC random forest.** LabEx Agro, Cemeb & Numev Joint Call 2016 (2018–2019). Rol: PI.
- **SelfAdapt: Adaptation to climate change under selfing — experimental test and selection footprints in *Medicago truncatula*.** Meta-programme INRA - Adapting Agriculture and Forestry to Climate Change 2012 (2013–2016). Rol: Co-coordinator. PI: L. Gay.

Publications

I have published 38 articles and 2 pre-prints. Selection of publications below.

```
library(scholar)
profileMNavascues <- get_profile("I9Rd7ocAAAAJ")
publicationsMNavascues <- get_publications("I9Rd7ocAAAAJ")
citation_historyMNavascues <- get_citation_history("I9Rd7ocAAAAJ")
par(mar=c(5,5,0,0)+0.2)
barplot(citation_historyMNavascues$cites,names.arg=citation_historyMNavascues$year,col="orange",box=)
```

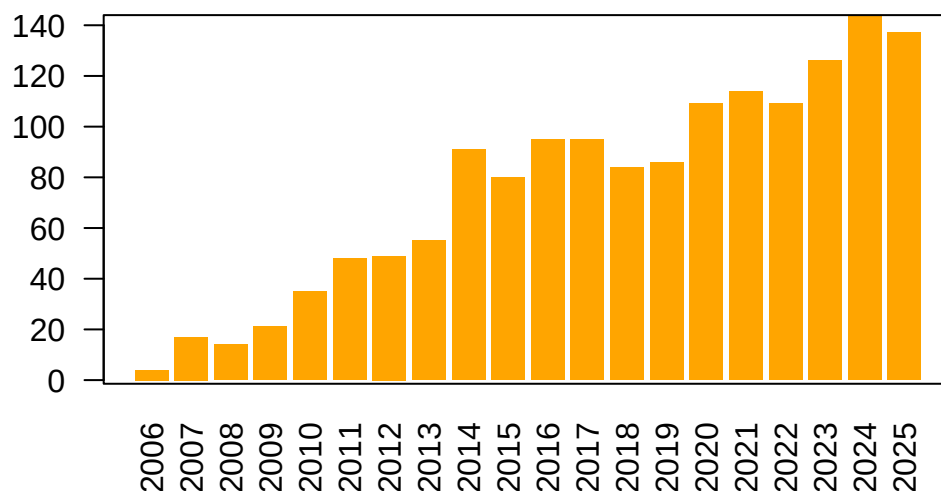


Figure 1: Number of citations per year according to Google Scholar.

- **SelNeTime: a python package inferring effective population size and selection intensity from genomic time series data.** M. Uhl, P. Bunel, *M de Navascués*, S. Boitard, B. Servin (2024) [doi:10.1101/2024.11.06.622284](https://doi.org/10.1101/2024.11.06.622284). [Open Peer-Review](#), [Open Code](#), [Open Code](#). Citations: 0.

- **Analysis of the abundance of radiocarbon samples as count data.** *M. de Navascués, C. Burgarella, M. Jakobsson* (2024) [doi:10.24072/pcjournal.522](https://doi.org/10.24072/pcjournal.522). [Open Peer-Review](#), [Open Code](#). Citations: [1](#).
- **Joint inference of adaptive and demographic history from temporal population genomic data.** V.A.C. Pavinato, S. De Mita, J.-M. Marin, *M. de Navascués* (2022) [doi:10.24072/pcjournal.203](https://doi.org/10.24072/pcjournal.203). [Open Peer-Review](#), [Open Data](#), [Open Code](#). Citations: [5](#).
- **Power and limits of selection genome scans on temporal data from a selfing population.** *M. de Navascués, A. Becheler, L. Gay, J. Ronfort, K. Loridon, R. Vitalis* (2021) [doi:10.24072/pcjournal.47](https://doi.org/10.24072/pcjournal.47). [Open Peer-Review](#), [Open Code](#), [Open Data](#). Citations: [7](#).
- **Demographic inference through approximate-Bayesian-computation skyline plots.** *M. de Navascués, R. Leblois, C. Burgarella* (2017) [doi:10.7717/peerj.3530](https://doi.org/10.7717/peerj.3530). [Open Code](#). Citations: [19](#).